Hanna Lachnitt

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Research Interests: Automated Reasoning, Formal Verification, Satisfiability Modulo Theories (SMT), Interactive Theorem Proving (ITP), Proof Certificates

PhD Research

expected graduation: summer 2027

My research has been centered around proof certificates for SMT solvers. My objective is to improve automation in the ITP Isabelle/HOL by reconstructing the SMT solver cvc5's proof certificates [3].

Education

Stanford University (USA) *since 09/2020*
PhD, Computer Science | Advisor: Clark Barrett | GPA: 4.0

TU Wien (Austria) *03/2018 – 06/2020*
MSc, Logic and Computation | Advisor: Laura Kovács | 1.0 (corresponds to 4.0 GPA), summa cum laude

Free University Berlin (Germany) *10/2014 – 09/2017*
BSc, Computer Science | 1.3 (corresponds to 3.7 GPA), summa cum laude

Research Experience

Visiting Researcher at UFMG (Brazil) *06/2023*
Supervisor: Haniel Barbosa | Publication: [6] | Parallel proof checking of an DRAT based SMT format

Visiting Researcher at Freiburg University (Germany) *09/2024*
Supervisor: Mathias Fleury | Improved the integration of CVC5 into Isabelle

Research Collaboration with Trippel Lab, Stanford University *11/2020 – 05/2022*
Supervisor: Caroline Trippel | Publication: [9] | Hardware Verification. Developed axiomatic security contracts.

Research Assistant, Automated Program Reasoning group, TU Wien *09/2019 – 05/2020*
Master's Thesis: Formalizing Graph Trail Properties
Supervisor: Laura Kovács | Publication: [10] | Formalizing Graph Trail Properties

Research Assistant, Formal Methods in Security group, TU Wien *06/2018 – 04/2019*
Supervisor: Matteo Maffei | Formalizing Ethereum smart-contracts in F*

Research Internship at the University of Cambridge (UK) *06/2019 – 09/2019*
Supervisors: Anthony Bordg, Lawrence Paulson | Publication: [13] | Formalizing quantum algorithms in Isabelle/HOL

Bachelor's Thesis *11/2016 – 01/2017*
Supervisor: Christoph Benzmüller | Systematic Verification of the Intuitionistic Modal Logic Cube in Isabelle/HOL

Awards and Honors

Academic Awards

Distinguished Artifact Reviewer Award *04/2026*
International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)

Rance Cleaveland Test-of-Time Tool Award *04/2026*
For CVC4/cvc5 at International Joint Conferences On Theory and Practice of Software (ETAPS)

VerifyThis Verification Competition *04/2024*
"Most impressive tool medley"

SCP Best Tool Paper Award *04/2022*
For *cvc5: A Versatile and Industrial-Strength SMT Solver* at ETAPS

Stipends

Siemens Award for Excellence *01/2020*
Master Logic and Computation: Awarded to the best female students of the academic year

Helmut Veith Stipend *03/2018 – 02/2020*
Financial scholarship for female students pursuing a Logic Master's program at TU Wien awarded to a single person per year

Deutschlandstipendium (German National Scholarship) *10/2015 – 09/2016*
Awarded by the Federal Republic of Germany for outstanding academic achievements

Invited Seminar Participation

Banff 26w5626 - Theory and Practice of SAT and Combinatorial Solving *01/2026*
Dagstuhl 25231 - Certifying Algorithms for Automated Reasoning *06/2025*
Dagstuhl 23471 - The Next Generation of Deduction Systems: From Composition to Compositionality *11/2023*

Teaching & Mentorship

Student Instructor at Stanford University *09/2025 – 12/2025*
CS257: Introduction to Automated Reasoning
Co-taught this class with Professor Caroline Trippel; gave half the lectures.

Teaching Assistant at Stanford University *09/2022 – 12/2022*
CS257: Introduction to Automated Reasoning *09/2023 – 12/2023*
Duties included office hours, correcting assignments, exams, and supervising the final class project.
Designed exercise sheets.

Mentoring *2022 – 2025*
Jibiana Jakpor (now a PhD student at MIT)

Service

External Service

Program Committee member of the 28th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC 2026) *2026*

Program Committee member of the Satisfiability Checking and Symbolic Computation (SC^2) Workshop *2026*

Reviewer CADE, TACAS, SC^2, SYNASC, Mathematics in Computer Science Journal *various*

Artifact reviewer TACAS, ATVA *2023, 2024, 2025*

Volunteering at conferences FloC, CAV *2018,2023*

Stanford Computer Science Department Service

Student application support program *10/2026*

Faculty Search Committee member *2025*

Chair of the PhD Advisory Council *11/2022 – 11/2023*

Co-Organizer of the Women in CS Lunch *09/2021 – 12/2022*

Student Reader on the Admissions Committee for PhD students *10/2022 – 12/2022*

Other:
Part of a team representing the department at the Grace Hopper Conference, student buddy for new PhD students, panel member during admit days, ...

Professional Experience

Apple *06/2026 – 08/2026*
Hardware Formal Verification Intern | Supervisor: Thomas Rasmussen

Amazon (AWS) *06/2022 – 09/2022*
Applied Science Intern | Supervisor: Robert Jones | Publication: [4] | Helped develop SMT proof format EDRAT.

Apple *06/2021 – 09/2021*
Software Formal Verification Intern | Supervisor: Tim Pruss | Verified parts of a kernel system in Isabelle/HOL.

Bundesdruckerei GmbH *08/2016 – 09/2016*
Intern | Contributed to ISO/IEC 39794-5 standard on face image data.

List of Publications

Conference Papers

[1] A. Reynolds, H.-J. Schurr, H. Barbosa, O. Israel, J. Jakpor, H. Lachnitt, A. Mohamed, A. Niemetz, M. Preiner, Y. Zohar, R. B. Jones, C. W. Barrett, and C. Tinelli. **The Cooperating Proof Calculus: Comprehensive Proofs for an SMT Solver**. In *38th International Conference on Computer Aided Verification (CAV 2026)*, Part II, LNCS 16683, pp. 188–212. Springer, 2026.

[2] S. Binder, H. Lachnitt, and K. Kosaian. **Apply2Isar: Automatically Converting Isabelle/HOL Apply-Style Proofs to Structured Isar**. In *17th International Conference on Interactive Theorem Proving (ITP 2026)*, LIPIcs 382, pp. 18:1–18:20. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026.

[3] H. Lachnitt, M. Fleury, H. Barbosa, J. Jakpor, B. Andreotti, A. Reynolds, H.-J. Schurr, C. W. Barrett, and C. Tinelli. **Improving the SMT Proof Reconstruction Pipeline in Isabelle/HOL**. In *16th International Conference on Interactive Theorem Proving (ITP 2025)*, LIPIcs 352, pp. 26:1–26:22. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025.

[4] S. Hitarth, C. R. Codel, H. Lachnitt, and B. Dutertre. **Extending DRAT to SMT**. In *Formal Methods in Computer-Aided Design (FMCAD 2024)*, pp. 1–11. IEEE, 2024.

[5] H. Lachnitt, M. Fleury, L. Aniva, A. Reynolds, H. Barbosa, A. Nötzli, C. W. Barrett, and C. Tinelli. **IsaRare: Automatic Verification of SMT Rewrites in Isabelle/HOL**. In *30th International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2024)*, pp. 311–330. Springer, 2024.

[6] B. Andreotti, H. Lachnitt, and H. Barbosa. **Carcara: An Efficient Proof Checker and Elaborator for SMT Proofs in the Alethe Format**. In *29th International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2023)*, Part I, LNCS 13993, pp. 367–386. Springer, 2023.

[7] H. Barbosa, A. Reynolds, G. Kremer, H. Lachnitt, A. Niemetz, A. Nötzli, A. Ozdemir, M. Preiner, A. Viswanathan, S. Viteri, Y. Zohar, C. Tinelli, and C. W. Barrett. **Flexible Proof Production in an Industrial-Strength SMT Solver**. In *11th International Joint Conference on Automated Reasoning (IJCAR 2022)*, LNCS 13385, pp. 15–35. Springer, 2022.

[8] H. Barbosa, C. W. Barrett, M. Brain, G. Kremer, H. Lachnitt, M. Mann, A. Mohamed, M. Mohamed, A. Niemetz, A. Nötzli, A. Ozdemir, M. Preiner, A. Reynolds, Y. Sheng, C. Tinelli, and Y. Zohar. **cvc5: A Versatile and Industrial-Strength SMT Solver**. In *28th International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS 2022)*, Part I, LNCS 13243, pp. 415–442. Springer, 2022.

[9] N. Mosier, H. Lachnitt, H. Nemati, and C. Trippel. **Axiomatic Hardware-Software Contracts for Security**. In *49th Annual International Symposium on Computer Architecture (ISCA 2022)*, pp. 72–86. ACM, 2022.

[10] L. Kovács, H. Lachnitt, and S. Szeider. **Formalizing Graph Trail Properties in Isabelle/HOL**. In *13th International Conference on Intelligent Computer Mathematics (CICM 2020)*, LNCS 12236, pp. 190–205. Springer, 2020.

Journal Papers

[11] (accepted) B. Andreotti, H. Lachnitt, and H. Barbosa. **An Efficient Proof Checker and Elaborator for SMT Proofs in the Alethe Format**. To appear in *International Journal on Software Tools for Technology Transfer (STTT)*, Special Issue for TACAS 2023 (by invitation only).

[12] H. Barbosa, C. W. Barrett, B. Cook, B. Dutertre, G. Kremer, H. Lachnitt, A. Niemetz, A. Nötzli, A. Ozdemir, M. Preiner, A. Reynolds, C. Tinelli, and Y. Zohar. **Generating and Exploiting Automated Reasoning Proof Certificates**. In *Communications of the ACM* 66(10), pp. 86–95. ACM, 2023.

[13] A. Bordg, H. Lachnitt, and Y. He. **Certified Quantum Computation in Isabelle/HOL**. In *Journal of Automated Reasoning* 65(5), pp. 691–709, 2021.

PRESENTATIONS

Selected Talks

Invited Talks on the *Reconstruction of cvc5 Proofs in Isabelle/HOL*:

DARPA PROVERS project, US *08/2025,09/2026*
LMU München, Germany *08/2026*
TU Wien, Austria *09/2024*
Freiburg University, Germany *09/2024*
Apple Inc., US *12/2023*

Other Invited Talks

UFMG, Brazil, Introduction into SMT solving *06/2023*

Conference and Workshops Talks

Improving the SMT Proof Reconstruction Pipeline in Isabelle/HOL, ITP *10/2025*
IsaRare: Automatic Verification of SMT Rewrites in Isabelle/HOL, TACAS *08/2024*
Automatic Verification of SMT Rewrites in Isabelle/HOL, SMT Workshop *07/2023*
Fine-Grained Reconstruction of cvc5 Proofs in Isabelle/HOL, FMCAD Student Forum *10/2022*
Dagstuhl Seminar 25231

Guest Lecture

ITP Introduction and Tutorial, *CS357S: Formal Methods for Computer Systems, Stanford University* *08/2026*
Isabelle/HOL Tutorial, *Automated Deduction, TU Wien* *05/2019*

Posters

ETAPCS Poster and Tool Demo *04/2024*
FMCAD Poster Session *10/2022*